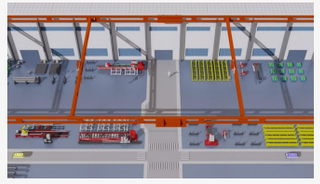
Digital manufacturing environment



Isaac Sim product line with single-axis disturbances

Human · Machine · Gantry · Buffer · Robot · Material

Input tensor (schematic)

27 features

38 resources

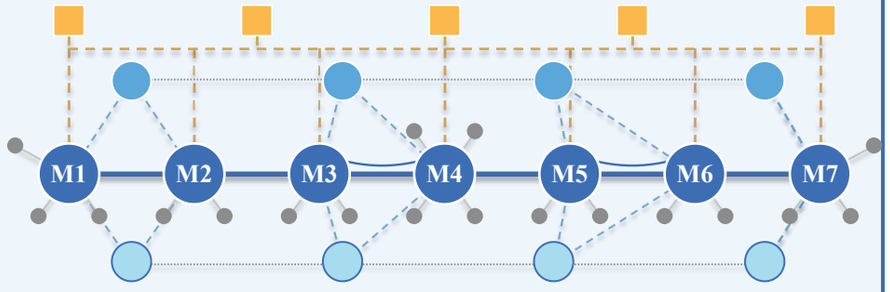
30 time windows

X_t: 30 × 38 × 27

Multi-entity heterogeneous graph

✗ **Prior work** — machine-only nodes

✓ **PHGST-DP** — 5 entity types, typed edges



machine · gantry · robot · human · buffer

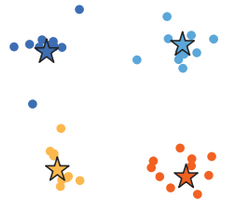
process · handling · weak · buffer link

Process-informed event labels

✗ **Prior work** — threshold spikes ≠ bottleneck

✓ **PHGST-DP** — sustained events, 8-min rule

Short-pattern clustering



16 pattern keys K^P

Type-specific rule b^{raw}

persistence ↓ m = 8, g = 1

Event target (w, s, d)

history · s · d

PHGST-DP

X_t

G, M^G, M^S

pattern keys K^P

training targets

Grouped embedding

Queue · Stall · Logistics · Shortage · Context · Type → mean

Encoder block × 5 (d = 80)

Delayed-propagation module

✗ **Prior work** — same-time links

✓ **PHGST-DP** — lagged precursors

Pattern attention — C^P added to geo. keys

- **Geographic attn ×2** mask M^G, keys + C^P
- **Temporal attn ×4** per resource, 30 windows
- **Semantic attn ×2** mask M^S, similar resources

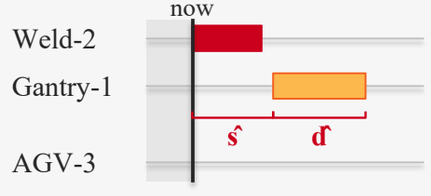
→ **Concat + project** → **Add & Norm + FFN**

Multi-task heads

- **Occurrence** p^{cont} / p^{onset}
- **Start** π^s · **duration** d^{ℓ}
- **Prefix** $\hat{p}$ · **occupancy** $\hat{O}$
- **Cause** $\hat{c}$ · **remaining** ℓ^{ℓ}

validation-controlled decoding

Bottleneck report



Process cause $\hat{c}$

Transport · Shortage · Starvation · Queue

Remaining order time ℓ^{ℓ}

12 min left